

1.14 Infinite Limits and Vertical Asymptotes

Calculus

Solutions

Practice

Identify the vertical asymptotes of each function.

$$1. f(x) = \frac{\cancel{x}^6}{x^2 - 9x + 18} \frac{(x-6)(x-3)}{(x-6)(x-3)}$$

$$X = 3$$

$$2. f(x) = \frac{2x^2 - x - 3}{3x^2 + 4x + 1} \frac{(2x-3)(x+1)}{(3x+1)(x+1)}$$

$$X = -\frac{1}{3}$$

$$3. f(x) = \frac{x^2 - x - 12}{x+7} \frac{(x-4)(x+3)}{(x+7)}$$

$$X = -7$$

$$4. f(x) = \frac{3x^2 - 11x + 10}{x-2} \frac{(3x-5)(x-2)}{(x-2)}$$

$$\text{no vertical asymptotes}$$

$$5. f(x) = \frac{x^3 + 2x^2 - 24x}{x^2 - x} \frac{x(x+6)(x-4)}{x(x-1)}$$

$$X = 1$$

$$6. f(x) = \frac{7x^2 + 4x - 3}{7x-3} \frac{(7x-3)(x+1)}{7x-3}$$

$$\text{no vertical asymptotes}$$

7. $f(x) = \csc(2x)$ on the interval $[0, \pi]$

$$f(x) = \frac{1}{\sin(2x)} \quad \sin(2x) = 0$$

$$2x = 0 \quad 2x = \pi \quad 2x = 2\pi \quad 2x = 3\pi$$

$$x = 0 \quad x = \frac{\pi}{2} \quad x = \pi$$

$$x = \frac{3\pi}{2}$$

8. $f(x) = \sec\left(\frac{x}{2}\right)$ on the interval $[-\pi, \pi]$

$$f(x) = \frac{1}{\cos\left(\frac{x}{2}\right)} \quad \cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = -\frac{\pi}{2} \quad \frac{x}{2} = \frac{\pi}{2} \quad \frac{x}{2} = \frac{3\pi}{2}$$

$$x = -\pi \quad x = \pi$$

$$x = 3\pi$$

Evaluate the limit.

9. $\lim_{x \rightarrow 1^+} \frac{x^2}{x-1} = \frac{(1.0001)^2}{1.0001-1}$

$$\approx \frac{1}{0.0001}$$

$$\infty$$

10. $\lim_{x \rightarrow -2^-} \frac{-3}{x+2} = \frac{-3}{-2.0001+2}$

$$\frac{-3}{-0.0001}$$

$$\infty$$

11. $\lim_{x \rightarrow 1^+} \frac{x-2}{x^2-3x+2} = \frac{x-2}{(x-2)(x-1)}$

$$\frac{1}{x-1} \rightarrow \frac{1}{1.0001-1}$$

$$\frac{1}{0.0001}$$

$$\infty$$

12. $\lim_{x \rightarrow -2} \frac{x+3}{x^2+4x+4} = \frac{x+3}{(x+2)(x+2)}$

$$x \rightarrow -2^- = \frac{-2.001+3}{(-2.001+2)(-2.001+2)} = \frac{0.999}{(-.001)(-.001)}$$

$$x \rightarrow -2^+ = \frac{-1.999+3}{(0.001)(0.001)} = \frac{1.001}{\text{small}} = \infty$$

$$\infty$$

13. $\lim_{x \rightarrow -1} \frac{x-1}{x^2-x-2} = \frac{x-1}{(x-2)(x+1)}$

$$x \rightarrow -1^- = \frac{-1.001-1}{(-1.001-2)(-1.001+1)} = \frac{-2.001}{\text{almost zero (positive)}} = -\infty$$

$$x \rightarrow -1^+ = \frac{-0.999-1}{(-0.999-2)(-0.999+1)} = \frac{-1.999}{\text{almost zero (negative)}} = \infty$$

No Limit!

14. $\lim_{x \rightarrow 3} -\frac{x^2}{3x-9}$

$$x \rightarrow 3^- \approx -\frac{8.999}{8.999-9} \approx -\frac{8.999}{-0.001} = \infty$$

$$x \rightarrow 3^+ \approx -\frac{9.001}{9.001-9} \approx -\frac{9.001}{0.001} = -\infty$$

No Limit

15. $\lim_{x \rightarrow -3} \frac{x-1}{x^2+6x+9} = \frac{(x-1)}{(x+3)(x+3)}$

$$x \rightarrow -3^- \approx \frac{-3.001-1}{(-0.001)(-0.001)} \approx \frac{-4.001}{0.000001} = -\infty$$

$$x \rightarrow -3^+ \approx \frac{-2.999-1}{(0.001)(0.001)} \approx \frac{-3.999}{0.000001} = -\infty$$

$$-\infty$$

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Test Prep

16. $\lim_{x \rightarrow 0^+} \frac{\cos x}{x} = \frac{\cos(0.00001)}{0.00001} \rightarrow \frac{\text{almost } 1}{\text{almost zero}} \rightarrow \infty$

E

(A) $-\infty$

(B) -1

(C) 0

(D) 1

(E) ∞

17. Consider the functions $f(x) = \frac{1}{x}, x \neq 0$, and $g(x) = x \sin \frac{1}{x}, x \neq 0$. Which of the following describes the behavior of f and g as $x \rightarrow 0$?

$\lim_{x \rightarrow 0} f(x) = \text{DNE}$

$\lim_{x \rightarrow 0} g(x) = (\text{almost zero}) \sin(\infty)$

$\lim_{x \rightarrow 0} g(x) = 0$

Sine is between -1 and 1 .

(A) $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{x \rightarrow 0} g(x) = 0$

(B) $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ do not exist.

(C) $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{x \rightarrow 0} g(x)$ does not exist.

(D) $\lim_{x \rightarrow 0} f(x)$ does not exist and $\lim_{x \rightarrow 0} g(x) = 0$

(E) $\lim_{x \rightarrow 0} f(x) = \infty$ and $\lim_{x \rightarrow 0} g(x) = 0$

18. The function h is defined by $h(x) = \left(\frac{x^2-x-20}{x+4}\right) \ln\left(\frac{x^2+10x+25}{x^2+5x}\right)$. At what values of x does the graph of h have a vertical asymptote?

$\frac{(x+4)(x-5)}{x+4} \ln\left[\frac{(x+5)(x+5)}{x(x+5)}\right]$

$(x-5) \ln\left(\frac{x+5}{x}\right)$

$\ln\left(\frac{x+5}{x}\right)^{x-5}$

Properties of logarithms

$x=0$ is a V.A.
but if $\ln\left(\frac{x+5}{x}\right)$ has a negative exponent, then $\ln\left(\frac{x}{x+5}\right)$ has a V.A. at $x=-5$.

(A) $x = -5$ only

(B) $x = 0$ only

(C) $x = -5$ and $x = 0$ only

(D) $x = -5, x = 0$ and $x = -4$